

GAINESVILLE, TX, USA

WMO: 722552

Lat: 33.651N

Lon: 97.197W

Elev: 839

StdP: 14.26

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 93929

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.9	23.7	4.3	7.1	29.4	9.4	9.2	36.4	27.0	47.3	24.9	50.7	7.2	340	0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.0	100.1	72.3	98.7	72.8	95.3	74.0	78.4	90.9	77.6	90.5	76.8	89.3	9.2	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.1	135.9	85.2	73.5	128.5	82.5	73.0	126.6	82.1	42.6	90.6	41.6	91.0	40.7	88.9	84.2

Extreme Annual Design Conditions

Extreme Annual WS							Extreme Annual Temperature								n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years							
					Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max						
(a)	(b)	(c)			(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)						
24.1	20.7	18.7	DB		12.5	103.2	3.9	4.4	9.6	106.4	7.3	109.0	5.1	111.4	2.3	114.7						
			WB		10.7	81.0	3.3	1.6	8.3	82.1	6.4	83.1	4.5	84.0	2.0	85.2						

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	64.1	43.4	47.0	56.3	62.7	71.0	79.7	83.2	83.7	76.4	65.1	54.7	44.8
	DBStd	16.32	9.88	10.87	9.76	7.78	7.05	4.64	4.28	4.88	5.82	8.65	9.16	9.27
	HDD50	758	250	171	52	9	0	0	0	0	0	6	56	214
	HDD65	2690	669	508	296	129	27	0	0	0	2	107	324	628
	CDD50	5905	47	87	247	390	652	890	1030	1045	793	474	198	52
	CDD65	2362	1	5	27	59	213	440	565	580	345	110	16	1
	CDH74	25974	13	45	189	443	1809	4695	6996	7228	3440	974	130	12
	CDH80	12838	1	6	28	78	621	2276	3882	4058	1570	295	23	0
Wind	WSAvg	8.1	8.3	9.1	9.9	10.0	8.7	7.9	6.7	6.3	6.3	7.4	8.3	8.1
Precipitation	PrecAvg	39.8	2.0	2.6	3.7	4.1	4.4	4.3	2.6	2.6	3.7	4.0	2.7	2.9
	PrecMax	53.5	4.6	8.0	8.0	12.0	9.3	15.9	8.2	8.0	10.9	11.3	7.3	8.5
	PrecMin	22.6	0.0	0.1	0.3	1.5	0.5	0.7	0.0	0.1	0.8	0.2	0.4	0.3
	PrecStd	8.7	1.1	1.9	2.0	2.7	2.2	3.3	2.2	1.8	2.4	3.0	2.2	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	75.2	79.2	82.6	88.0	93.5	99.1	105.5	107.8	99.0	90.3	82.5	75.2
		MCWB	55.1	60.9	64.9	66.1	73.1	73.5	72.2	71.7	71.8	70.4	64.8	59.2
	2%	DB	70.3	73.4	79.1	81.4	90.2	95.4	99.8	100.4	94.6	87.7	78.5	69.9
		MCWB	56.6	59.6	64.2	66.5	72.8	74.1	72.0	72.5	72.0	70.6	62.7	57.3
	5%	DB	65.3	69.6	75.0	79.1	87.6	93.3	98.5	99.1	91.3	82.8	73.3	64.2
		MCWB	52.9	58.7	62.1	65.5	73.2	73.6	72.8	73.0	72.0	67.8	61.4	54.7
	10%	DB	60.7	64.4	72.6	75.6	82.9	91.1	95.0	95.5	89.0	79.4	70.0	60.5
		MCWB	50.4	56.0	61.3	63.9	70.6	74.9	74.2	74.4	71.6	65.7	60.3	52.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.5	66.5	70.9	73.3	78.2	80.3	80.0	79.6	78.2	75.2	69.6	65.5
		MCDB	68.2	73.7	76.9	80.4	89.7	91.7	92.1	91.0	90.2	85.1	77.3	71.6
	2%	WB	59.6	64.0	68.1	71.2	76.2	78.2	78.3	78.1	76.3	73.3	67.1	61.3
		MCDB	64.6	70.1	74.0	77.5	86.2	90.3	91.1	91.2	87.8	82.3	73.5	66.2
	5%	WB	55.5	60.9	66.2	69.2	74.2	77.2	77.3	77.2	75.1	71.7	64.8	57.6
		MCDB	62.8	66.3	73.1	75.2	82.7	89.1	90.2	90.8	85.7	78.8	70.8	61.5
	10%	WB	51.0	57.1	63.9	67.1	72.5	76.0	76.4	76.3	73.9	69.9	62.0	53.2
		MCDB	58.8	64.2	70.5	73.9	80.2	87.2	89.0	89.4	83.9	76.6	68.7	58.3
Mean Daily Temperature Range	5% DB	MDBR	22.1	21.0	21.5	21.1	19.3	19.3	20.9	21.0	20.8	21.9	21.9	20.5
		MCDBR	30.5	28.4	25.0	23.2	22.9	21.8	24.9	24.9	23.9	24.5	25.9	27.7
		MCWBR	18.6	17.3	14.0	13.2	10.4	8.0	6.8	6.6	8.4	11.5	15.1	17.2
	5% WB	MCDBR	25.6	23.9	19.4	18.6	18.9	19.5	20.3	21.0	19.2	19.1	21.2	21.5
		MCWBR	19.1	17.5	13.1	13.2	10.6	8.8	7.8	7.2	8.2	10.6	15.8	16.9
Clear-Sky Solar Irradiance	taub	0.307	0.317	0.351	0.381	0.419	0.432	0.434	0.433	0.392	0.352	0.326	0.311	
	taud	2.498	2.482	2.408	2.346	2.304	2.306	2.306	2.336	2.318	2.382	2.479	2.490	2.488
	Ebn at Noon	289	297	293	287	276	271	270	269	276	281	279	280	
	Edh at Noon	27	30	36	40	42	42	40	40	36	30	27	26	
All-Sky Solar Radiation	RadAvg	902	1076	1415	1723	1859	2095	2128	1954	1627	1266	962	785	
	RadStd	93	164	102	138	190	207	138	117	154	187	124	95	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (47 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page